

Malignant spinal cord compression in cerebral glioblastoma multiforme: a multicenter case series and review of the literature.

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Glioblastoma multiforme (GBM) is the most common malignant primary brain tumor in adults. Compared with other malignancies, remote metastases in GBM are rare. However, multicentric spreading within the central nervous system is common and also metastases to the spinal cord have been reported. Some of these drop metastases may also lead to malignant spinal cord compression (MSCC). We retrospectively identified nine patients from 2001 to 2010 and performed data analysis according to a standardized clinical protocol. We also provide a review of the literature on this rare condition. MSCC from cerebral GBM is rare and is found in approximately 1 % of GBM patients. Median age of 54 years in this case series is comparable with that of GBM patients without MSCC. Treatment regimens for cerebral GBM and overall survival was similar to those for patients without MSCC. Spinal metastasis seems to occur in the advanced state of the disease, and the outcome subsequently is extremely poor. All patients presented with multicentric radiological features of GBM on cerebral MRI when MSCC was diagnosed. Subependymal enhancement is another common radiological finding in GBM patients with spinal drop metastases. Steroids and focal radiotherapy were used to treat all patients, with little clinical benefit. This study is the largest case series of MSCC from cerebral GBM. Multicentric cerebral distribution and subependymal enhancement of GBM are observed on cerebral MRI at the time of MSCC. On the basis of our results, no specific treatment recommendations for MSCC in GBM patients can be given. However, accurate diagnosis of MSCC in GBM patients with spinal signs and symptoms can lead to adequate management of symptoms and improvement of quality of life in terms of best palliative care.

A case of soft tissue metastasis from glioblastoma and review of the literature.

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Despite recent advances, glioblastoma (GBM) remains incurable but is typically characterized by local tumor recurrence in the brain. Soft tissue metastases from GBM are extremely rare, with only eight reported cases in the literature. It has been postulated that this type of metastases rarely occurs due to inability of tumor cells to survive outside the brain milieu, physical barriers preventing tumor spread, and/or early patient death from this aggressive tumor. Here we present another case of soft tissue metastases that occurred in a young woman with GBM and review the previous literature.

[Spinal epidural metastasis of glioblastoma multiforme: a case report].

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It is generally accepted that the metastases of intracranial glioma to extracranial location are rare. In such a case the minimal criteria proposed originally by Weiss should usually be satisfied if a report is to be considered as an acceptable case of metastasizing central nervous system glioma outside the central nervous system. We report a case of glioblastoma multiforme, fulfilling Weiss' criteria, metastasizing the spinal epidural space. The patient was a 32-year-old male, who underwent craniotomy and subtotal removal of a glioblastoma multiforme in the left parietooccipital area. He was additionally treated with irradiation and chemotherapeutic agents. Twelve months after the craniotomy, he was admitted again to

our clinic because of sudden onset of severe lumbago, paraplegia and urinary disturbance. Diagnosis of a spinal epidural tumor was made and laminectomy (Th10-L1) was performed. At operation, an epidural mass was found, however no invasion to the spinal cord or dura was noted. Histological diagnosis of the tumor was glioblastoma multiforme. Although he was treated with radiation, pulmonary metastasis was manifested one month later, and the condition of the patient deteriorated. He died 21 months after the first operation and 8 months after the second operation. Even at the terminal stage, his consciousness was clear without any sign for recurrence of intracranial tumor. The general autopsy was done and multiple metastatic lesions of glioblastoma multiforme in paratracheal and paraaortic lymph node, left pleura, both lungs and spinal cord were observed. The present case suggests that the surgical intervention, irradiation, and chemotherapy may contribute to extracranial metastasis of a glioblastoma.

Extracranial metastases of a glioblastoma multiforme to the pleura, small bowel and pancreas.

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Extracranial metastases of glioblastoma multiforme (GBM) are rare and usually occur in the context of recurrent intracranial GBM. We present a 39-year-old man with histologically confirmed GBM. The patient remained well for nearly 2 years, with no signs of recurrent tumour. He then presented with distant recurrence within the brain at the same time as developing pneumonia and epigastric pain. A computed tomography scan of the patient's abdomen and chest showed several intra-abdominal masses, including one in the head of the pancreas as well as a separate mass at the base of the left lung. A computed tomography-guided biopsy of the pancreatic mass demonstrated histological appearances identical to those of the original GBM. This unusual case raises the possibility of a link between prolonged survival with GBM and the occurrence of extracranial disease.

Extracranial growth of glioblastoma multiforme.

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We present a 59-year-old woman who noted an enlarging lump on her forehead 6 months after a left frontotemporal craniotomy for tumor resection and chemoradiation of her primary glioblastoma multiforme (GBM). GBM is a highly aggressive intracranial neoplasm associated with the shortest survival time of any primary central nervous system malignancy. Extracranial metastasis is rare, especially without previous surgical disruption of the dura and calvarium, which has been postulated to cause seeding of the extracranial space with tumor cells. This patient's MRI revealed tumor recurrence for which she underwent repeat resection. Histopathology confirmed GBM with unmethylated O-6-methylguanine-DNA methyltransferase and wildtype isocitrate dehydrogenase 1 status, as well as tumor invasion through the bone and subdermal space. The genetic and molecular factors that predict extracranial invasion remain unclear and require further investigation. Emerging data on circulating tumor cells in GBM patients indicate that extraaxial metastasis may be part of the disease course in some, particularly in long term survivors. Furthermore, the proximity of calvarial and scalp lesions to previous surgical sites and the time course in which they emerge after surgery suggests that iatrogenic seeding may also play a role in metastasis. With heightened awareness of the phenomenon, surgical strategies such as watertight approximation of the dura, bone flap replacement, or changing surgical

instruments once the intradural component is complete may prove useful to prevent seeding. Prophylactic craniospinal irradiation may also be an appropriate tool in patients at high risk for metastasis, although this population is difficult to identify.

Single-port laparoscopic salpingectomy for the surgical treatment of ectopic pregnancy.

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To evaluate the feasibility of a single-port laparoscopic salpingectomy in the surgical treatment of tubal pregnancy. Prospective cohort study (Canadian Task Force classification II-2). University teaching hospital. Twenty women with tubal pregnancy, as determined by ultrasonography. All patients have undergone single-port laparoscopic salpingectomy. Entry through a single port was established with a wound retractor as fascial retractor and a surgical glove, which served as the working channels for the laparoscopic equipment. A 30-degree laparoscope and a rigid or flexible grasper were used during the procedure. Single-port laparoscopic salpingectomy was successfully performed in all 20 patients with ectopic pregnancy. The median operative time was 55 minutes (range 25-85 minutes), and blood loss in all patients was minimal. The median difference between preoperative and postoperative hemoglobin was 1.8 g/dL (range 0-3.2 g/dL). The median postoperative hospital stay was 2 days (range 2-4 days). No complication was encountered, nor was there any need for conversion to conventional laparoscopy. Single-port laparoscopic salpingectomy is feasible and promising. However, for drawing the definite conclusion of the surgical efficacy, additional investigations to compare this approach with conventional laparoscopy are needed.

The Trend, Feasibility, and Safety of Salpingectomy as a form of Permanent Sterilization.

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To assess the change in the rate of laparoscopic salpingectomy for sterilization after the release of the November 2013 Society of Gynecologic Oncology Clinical Practice Statement and the January 2015 American College of Obstetricians and Gynecologists Committee Opinion: Salpingectomy for Ovarian Cancer Prevention. We hypothesized there would be an increase in salpingectomy as a percentage of total laparoscopic sterilizations performed without an increase in complications when compared with conventional bilateral tubal ligation (BTL). A retrospective cohort study. Four university-affiliated hospitals in Houston, TX, and New York, NY. All women 21 years or older who underwent interval laparoscopic permanent sterilization between April 2013 and September 2016. Sterilization by bilateral salpingectomy or conventional tubal ligation. There were 454 sterilization procedures identified; 60% were BTLs, whereas 40% were salpingectomies. The rate of use of salpingectomy significantly increased from 5% to 9% in 2013 to 2014 to 78% by 2016. There was no significant difference in intraoperative or postoperative complications or estimated blood loss. The mean procedure time was 54 minutes for salpingectomy compared with 45 minutes for BTL ($p < .0001$). Salpingectomy was more likely to require 3 ports compared with 2 ports for BTL ($p < .0001$). The Society of Gynecologic Oncology and the American College of Obstetricians and Gynecologists' support of salpingectomy for ovarian cancer prevention increased its use for sterilization. Based on this study, laparoscopic bilateral salpingectomy is a safe method of sterilization without an increase in perioperative risk compared with conventional tubal

ligation. Physicians should incorporate these findings and implications when counseling patients regarding contraception and permanent sterilization.

Tips and Tricks for Performing Salpingectomy at the Time of Laparoscopic Hysterectomy.

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To demonstrate various techniques to perform salpingectomy efficiently at the time of laparoscopic hysterectomy. Step-by-step explanation of the techniques by video with narration (educational video) (Canadian Task Force Classification III). Salpingectomy at the time of laparoscopic hysterectomy. Ovarian cancer is the deadliest gynecologic malignancy and has no effective screening strategies for average-risk women. After recognizing that the origin site for pelvic serous carcinomas may be the fallopian tube, the Society of Gynecologic Oncology published a practice statement in November 2013 addressing the role of salpingectomy at the time of hysterectomy or other pelvic surgery in average-risk women. (<https://www.sgo.org/clinical-practice/guidelines/sgo-clinical-practice-statement-salpingectomy-for-ovarian-cancer-prevention>). They now recommend that these women consider opportunistic salpingectomy to reduce their risk of fallopian tube and ovarian cancers. Various techniques allow the surgeon to complete the salpingectomy in a highly efficient manner. Salpingectomy at the time of laparoscopic hysterectomy or other pelvic surgery should be considered in women at average risk of ovarian cancer. Salpingectomy can be performed either before or after control of the uterine blood supply. The surgical approach must also consider the coexisting pelvic pathology. Efficient dissection occurs if the surgeon maximizes exposure to the fallopian tube, optimizes presentation of the tissue to the working instrument, and provides gentle yet constant traction with accompanying countertraction. The fallopian tube specimen should be removed immediately to prevent its loss in the pelvis.

Safety outcomes of female sterilization by salpingectomy and tubal occlusion.

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Compare immediate and short-term complications and surgical times among women having laparoscopic salpingectomy or tubal occlusion for female sterilization. We used billing data to identify women having laparoscopic sterilization at our training institution between July 1, 2011, and June 30, 2015. We performed a retrospective chart review to extract demographic information, surgical times and complications within 30 days, including unscheduled clinic or emergency room visits. We categorized complications as immediate (prior to discharge) and short-term (within 30 days after the procedure). Surgeries including additional procedures other than IUD removal were considered mixed operations. Mixed operations and unilateral sterilization procedures were only included in safety evaluations. The 149 procedures included 81 salpingectomies (including 18 mixed operations and 2 unilateral salpingectomies) and 68 tubal occlusions (including 8 mixed operations). All procedures involved Obstetrics and Gynecology residents. Salpingectomy and occlusion procedures had similar immediate (2.5% vs. 2.9%, $p=1.0$) and short-term (4.9% vs. 14.7%, $p=.051$) complication rates. Surgical time averaged 6 min longer for salpingectomies than occlusion procedures (44 vs. 38 min, respectively, $p=.018$). Average surgical times were shorter with more experienced (3rd/4th year) residents than less experienced (1st/2nd year) residents for both salpingectomy (32 ± 18 min vs. 46 ± 13 min, respectively, $p=.124$) and occlusion

procedures (32 ± 13 min vs. 41 ± 12 min, respectively, $p = .026$). Salpingectomy for female sterilization takes slightly longer to complete than tubal occlusion procedures without evidence that it increases complications. Laparoscopic salpingectomy is a safe alternative to tubal occlusion with only a small increase in surgical time. Because salpingectomy offers higher efficacy and more ovarian cancer protection than occlusion procedures, salpingectomy should be an option offered to women seeking laparoscopic sterilization.

A Case of Uterine Lymphangioleiomyomatosis Complicated by Tuberous Sclerosis Complex.

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Lymphangioleiomyomatosis (LAM) is one of the presentations of perivascular epithelioid cell neoplasm that is frequently complicated by tuberous sclerosis complex (TSC). Here, we report an uncommon case of uterine LAM treated with everolimus, which is a mechanistic target of rapamycin (mTOR) inhibitor. A 42-year-old female patient (gravida 0) with a history of TSC presented with abdominal pain. Pelvic magnetic resonance imaging showed multiple masses in the uterine myometrium, suggesting tumors that may contain internal hemorrhagic components. The lesions were suspected as the root cause of her symptoms. After everolimus was administered for a previously diagnosed renal angioliipoma, her uterine tumors temporarily decreased in size. Subsequently, laparoscopic hysterectomy and bilateral salpingectomy were performed since she could not tolerate everolimus for a long period due to the medication's side effects. Furthermore, the patient was diagnosed with LAM through histopathological examination after surgical resection. Therefore, it is advisable to suspect and investigate uterine LAM when a patient with a history of TSC presents with irregular genital bleeding or abdominal pain. Moreover, mTOR inhibitors may be a treatment option, in addition to surgery, in cases of uterine LAM exacerbation.
